Test 2 Review Practice — Answers

MATH 130 · October 26, 2026 · Check your setup, not just the final number.

A1 $A = 5000(1 + 0.045/12)^{72} \approx$ **$6,546.52**

A2 $A = 5000e^{0.27} \approx$ **$6,549.82**; continuous earns about **$3.31** more.

A3 (a) $2400(1 + 0.30) =$ **$3,120.00** (b) $2400(1.015)^{20} \approx$ **$3,232.45**; **$112.45** more.

B1 (a) $3^4 = 81$ (b) $\log_2(1/8) = -3$

B2 (a) 5 (b) −3 (c) 5 (d) 1/2

B3 (a) $x = e^{2.4} \approx 11.0232$ (b) $x = 10^{-1.5} \approx 0.0316$

B4 (a) $x = \ln 42 / \ln 5 \approx 2.3223$ (b) $e^{0.2x} = 9$, $x = \ln 9 / 0.2 \approx 10.9861$

B5 $\ln 50 / \ln 7 \approx 2.0104$

C1 (a) $3x + 10 = 5x - 30$, $x = 20$; both 70° (b) $5x + 55 = 180$, $x = 25$; 65° and 115°

C2 (a) 63 m² (b) ½·7·(10 + 16) = 91 in²

C3 $s = 12$; $\sqrt{(12 \cdot 5 \cdot 4 \cdot 3)} = \sqrt{720} = 12\sqrt{5} \approx 26.83$ ft²

C4 12·5 + 7.5·3 = 60 + 22.5 = **82.5 ft²** (or 12·8 − 4.5·3)

C5 (a) scale 10/6, EF = 15 (b) F = C = 180 − 48 − 67 = 65°

C6 8/12 = 10/DE, DE = 15

C7 leg = $\sqrt{(17^2 - 8^2)} = 15$ cm; area = ½·8·15 = 60 cm²

D1 47 + 23/60 + 42/3600 = 47.3950°

D2 0.4675·60 = 28.05′; 0.05·60 = 3″ → 128°28′3″

D3 (a) 5π/4 (b) 210° (c) 160.43°

D4 (a) 7π ≈ 21.99 cm (b) 31.5π ≈ 98.96 cm²

D5 θ = 15/6 = 2.5 rad ≈ 143.24°

D6 (a) 55° (b) ½(64 + 98) = 81°

D7 (a) C = 14π in, A = 49π in² (b) $\pi(13^2 - 10^2) = 69\pi \approx 216.77$ ft²

D8 (a) ω = 300·2π = 600π ≈ 1884.96 rad/min (b) v = 600π·13 = 7800π in/min; ×60 ÷ 63,360 ≈ 23.20 mph

D9 (a) $V = \pi \cdot 4^2 \cdot 10 = 160\pi \approx 502.65$ cm³ (b) $2\pi \cdot 4^2 + 2\pi \cdot 4 \cdot 10 = 112\pi \approx 351.86$ cm²